Cognitive Warrant

A Theoretical Framework for Evaluating Semantic Alignment in Knowledge Organization Systems

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"Knowledge domains are structured hierarchically. Some believe that this may not be a quirk of custom or chance but inevitable. Our brains are hardwired to perceive hierarchical relationships, and, consequently, the only way to comprehend a knowledge domain is through the structure they provide."

Elaine Svenonius

The Intellectual Foundation of Information Organization (2000)

The Cost of Misalignment

A patron's information need encounters a system built on a different cognitive architecture

1 Query Formulation

The patron arrives with a meaning — graded and contextual. The system requires a term. Even a terminologically accurate query may retrieve the wrong semantic neighborhood.

2 Results Navigation

Subject heading facets require the patron to map an unfamiliar category label onto their internal concept. When the KOS boundary does not correspond to a cognitively coherent category, the facet degrades from navigation aid to interpretive error.

3 Query Reformulation

The patron attempts to reverse-engineer a categorical structure that does not share their organization. Lateral drift toward broader terms. Premature abandonment. Satisficing.

4 Invisible Failure

Transaction logs record that a query was issued and results were returned. The behavioral signature of cognitive misalignment — the gap between the patron's meaning and the system's structure — goes unmeasured.

What Warrant Has Not Asked

A century of evaluation — and still one question unanswered

Literary Warrant

Hulme, 1911

Asks:

When should a heading be created?

Answers:

When sufficient documents on a topic accumulate in the published literature.

Scope:

Archival and descriptive. Governs when to create a heading — not whether its categorical placement is cognitively meaningful.

User Warrant

20th–21st c.

Asks:

What terms do users actually use?

Answers:

Headings should reflect the vocabulary of information seekers.

Scope:

Lexical and behavioral. Operates at the level of surface terminology — not deep semantic processing or categorical organization.

Cognitive Warrant

This study

Asks:

Do categorical boundaries correspond to cognitive reality?

Answers:

A boundary has cognitive warrant if it corresponds to a measurable discontinuity in human semantic processing, above and beyond continuous similarity.

Scope:

Empirical and falsifiable. Treats KOS as testable models of conceptual structure.

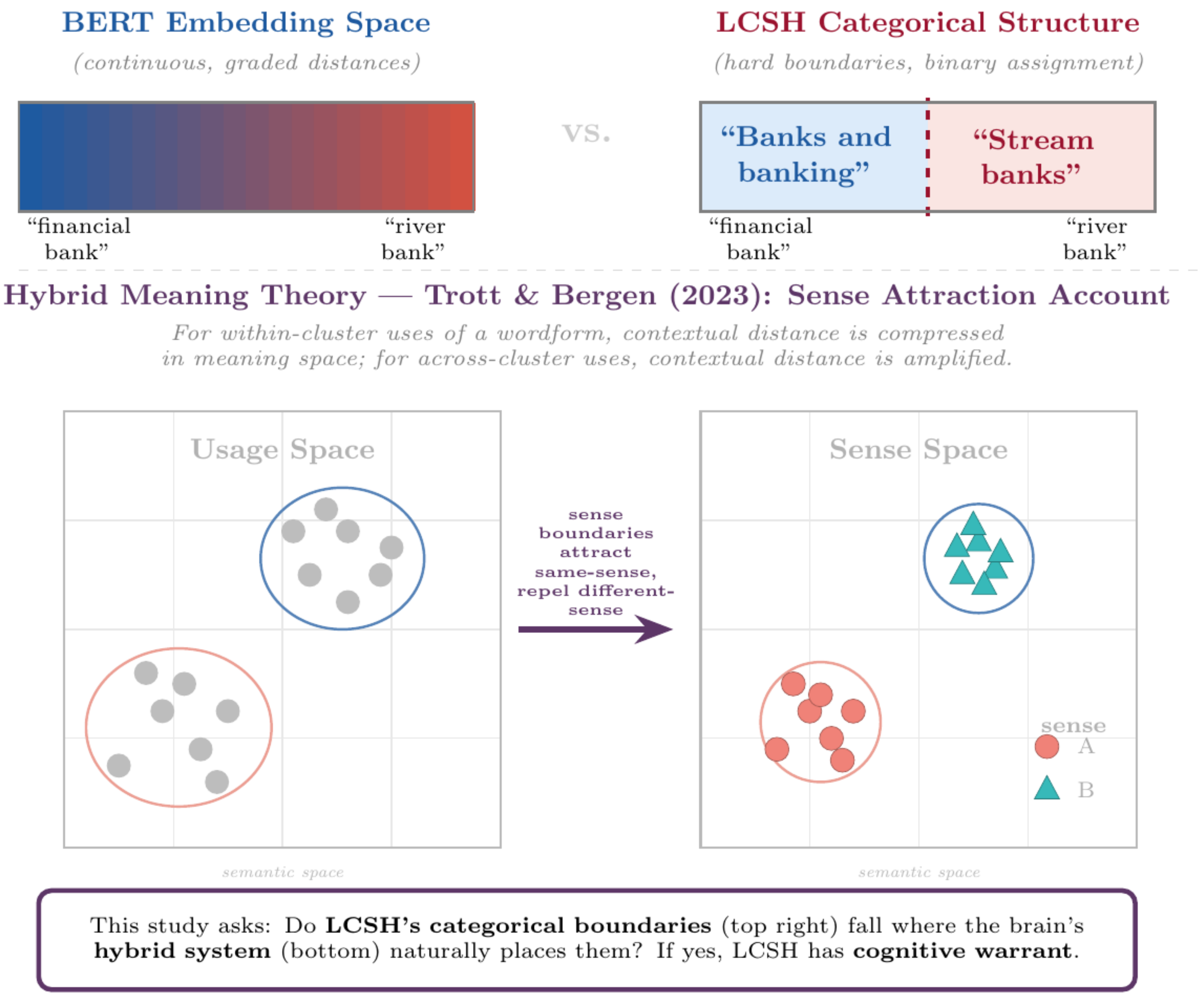
Cognitive Warrant

The extent to which a knowledge organization system reduces processing difficulty by resolving uncertainty in semantic interpretation.

Not when to create a heading. Whether its categorical placement is psychologically valid.

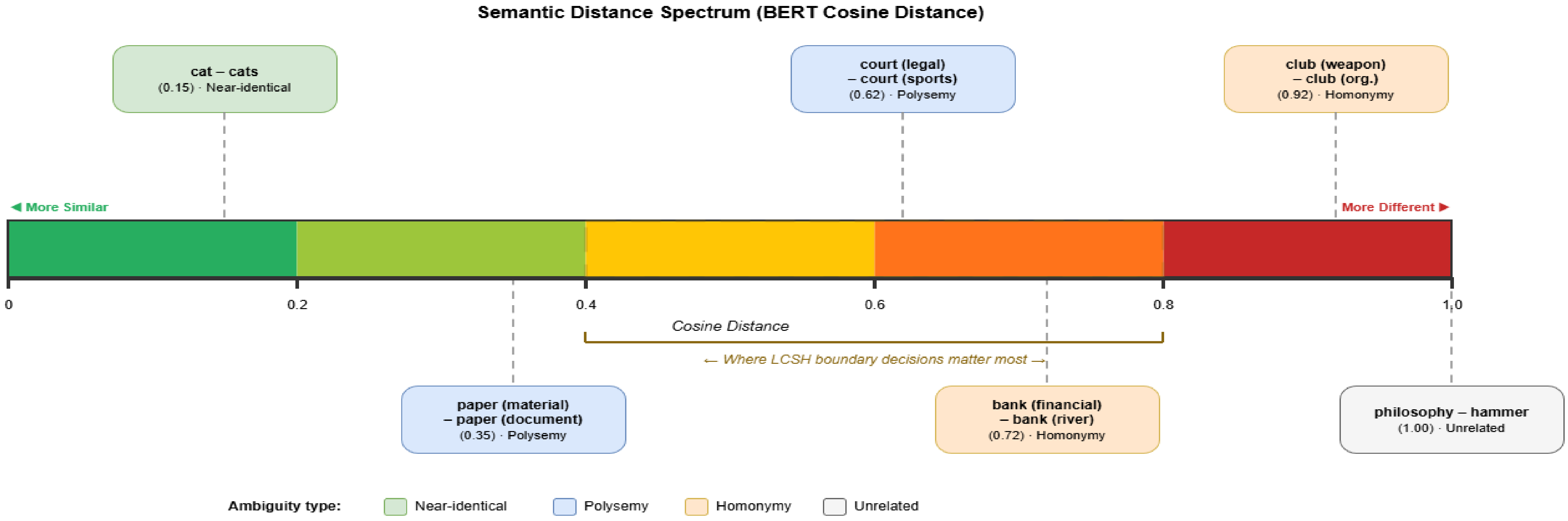
Theoretical foundations

BERT, LCSH, and Hybrid Meaning Theory



Semantic distance spectrum

Illustrating possible values of BERT cosine distance between word pairs



Cognitive Warrant Treats KOS as Falsifiable Models

KOS evaluation moves from qualitative consistency to quantitative information yield

Formal Definition

A categorical boundary in a KOS has cognitive warrant if and only if it corresponds to a measurable discontinuity in human judgments of semantic similarity or processing difficulty, above and beyond what is predicted by continuous semantic distance alone.

74.5% of variance in BERT distance is unaccounted for by categorical boundary status (Trott & Bergen, 2023). The two predictors carry independent information.

Empirically falsifiable

KOS categorical structure makes testable predictions about human behavior. If $CI(d) = 0$, the system adds no cognitive value. If $CI(d) < 0$, it actively misleads.

Operationally precise

pCI identifies specific headings as revision priorities — not broad characterizations of system failure but item-level diagnostic scores.

Cross-system comparable

$\overline{CI}$ and η are estimable from behavioral data and directly comparable across any KOS with categorical boundary structure.

H3 Is a Mathematical Consequence, Not a Prediction

If boundaries provide any useful information, their effects must concentrate at intermediate distances

Conditional Entropy

$$H(Y \mid d) = -\sum_{y\in\{0,1\}} P(y \mid d) \log P(y \mid d)$$

Residual uncertainty after conditioning on distance.
Peaks at intermediate *d*.

Cognitive Information

$$\mathrm{CI}(d) = H(Y \mid d) - H(Y \mid d, B) \equiv I(Y; B \mid d)$$

Reduction in uncertainty about sense identity Y provided by boundary *B*, given distance *d*.

Chain Rule Decomposition

$$I(Y; d, B) = \underbrace{I(Y; d)}_{\text{continuous}} + \underbrace{I(Y; B \mid d)}_{\mathrm{CI}(d):\ \text{categorical}}$$

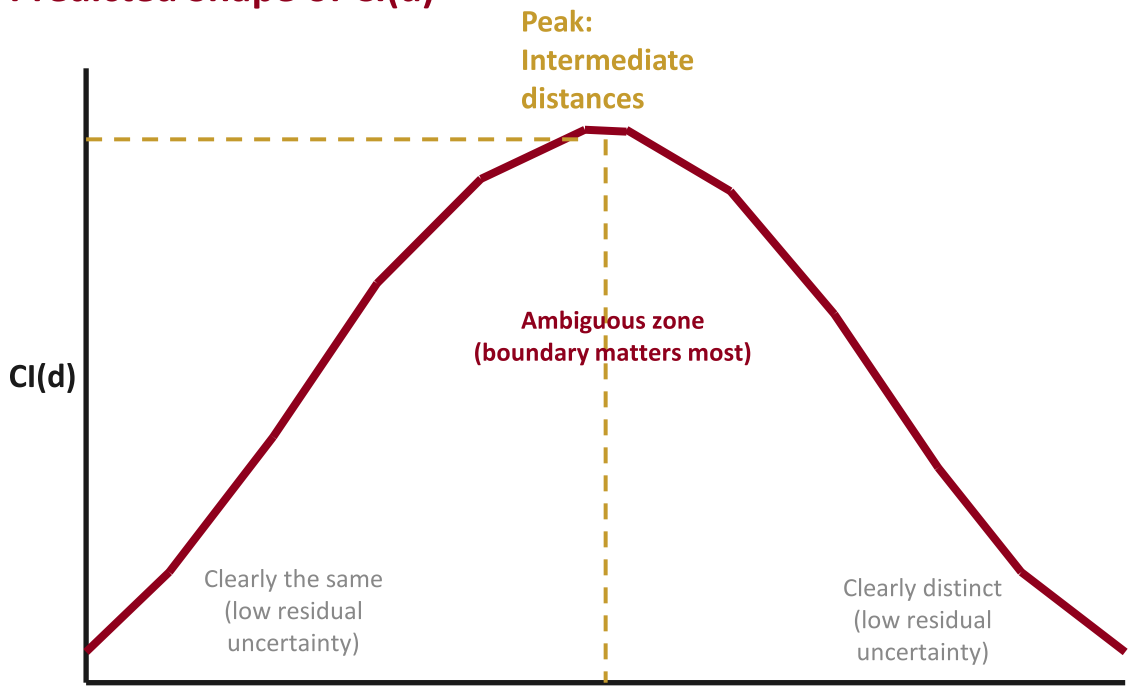
Continuous + categorical contributions to disambiguation.
Parallels Trott and Bergen's (2023) hybrid meaning theory.

H3 as Theorem

$$0 \le \mathrm{CI}(d) \le H(Y \mid d) \rightarrow \text{peak at interior } d$$

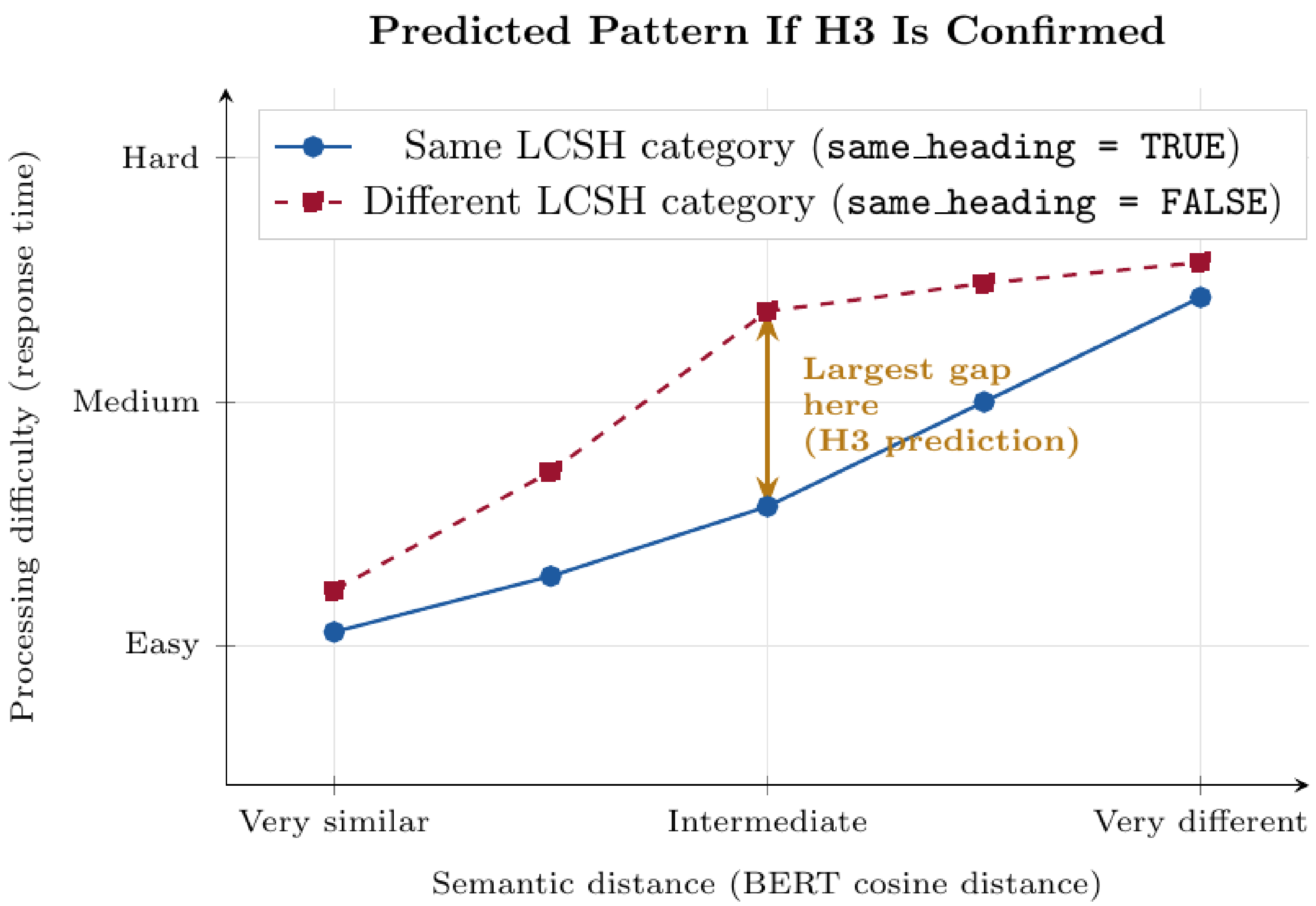
If H2 holds, CI must achieve its maximum at intermediate distances.
Proven from the properties of binary entropy.

Predicted Shape of CI(d)



The regression test of H3 is not asking whether the peak exists in principle — the mathematics guarantees it must if H2 holds.
It is asking whether the peak is detectable at the study's sample size.

H3: Boundary effects peak in the “ambiguous zone” of semantic distance



A Family of Evaluation Metrics — With Real Results

The framework derives quantitative tools applicable to any KOS with categorical boundary structure

$\overline{\text{CI}}$ Expected Cognitive Information

System-level score. Average uncertainty reduction provided by categorical structure, weighted by the empirical distribution of semantic distances.

Estimable from behavioral data. Additive across independent systems. Directly comparable across KOS evaluated against the same stimuli.

pCI Pointwise Cognitive Information

Item-level diagnostic. The log-ratio of how probable the correct sense was under the full model vs. the baseline — for each specific heading assignment.

pCI < 0: heading actively misleads sense identification. Ranked list = empirically prioritized revision agenda.

η Warrant Efficiency

Proportion of residual sense uncertainty (after conditioning on d) that LCSH boundaries resolve.

Range [0,1]. Normalized — enables cross-system comparison independent of baseline entropy magnitude.

Preliminary Results — 150 Stimulus Pairings

$\overline{\text{CI}}$ = 0.1299 bits	*Average uncertainty reduction above BERT distance*
η = 18.0%	*Residual sense ambiguity resolved by LCSH boundaries*
H2 LRT = $p < 0.0001$	*LCSH boundaries carry independent cognitive signal*

pCI Revision Priority Ranking:

dinner	*Polysemy*	−2.45	*Same heading; different senses*
film	*Polysemy*	−0.63	*Diff heading; same sense*
course	*Polysemy*	+6.15	*Correct — preserves cognitive value*
drill	*Homonymy*	+5.69	*Correct — preserves cognitive value*
tape	*Polysemy*	+4.91	*Correct — preserves cognitive value*

An subject authority record in LCSH

The LCSH mapping and graph construction are based on information found here

AUTHORIZED HEADING: Cognitive psychology
id.loc.gov/authorities/subjects/sh85026984

SN	Here are entered works on the branch of psychology that studies mental processes such as perception, memory, and reasoning.
LCC	BF311 (Library of Congress Classification shelf mark)
UF	Mental processes (if someone searches these, they are redirected here)
UF	Cognition—Psychological aspects
BT	Psychology (the broader category this heading belongs to)
NT	Attention (more specific topics that live inside this one)
NT	Memory; Perception; Problem solving (. . . and more)
RT	Neuropsychology (related, but neither broader nor narrower)

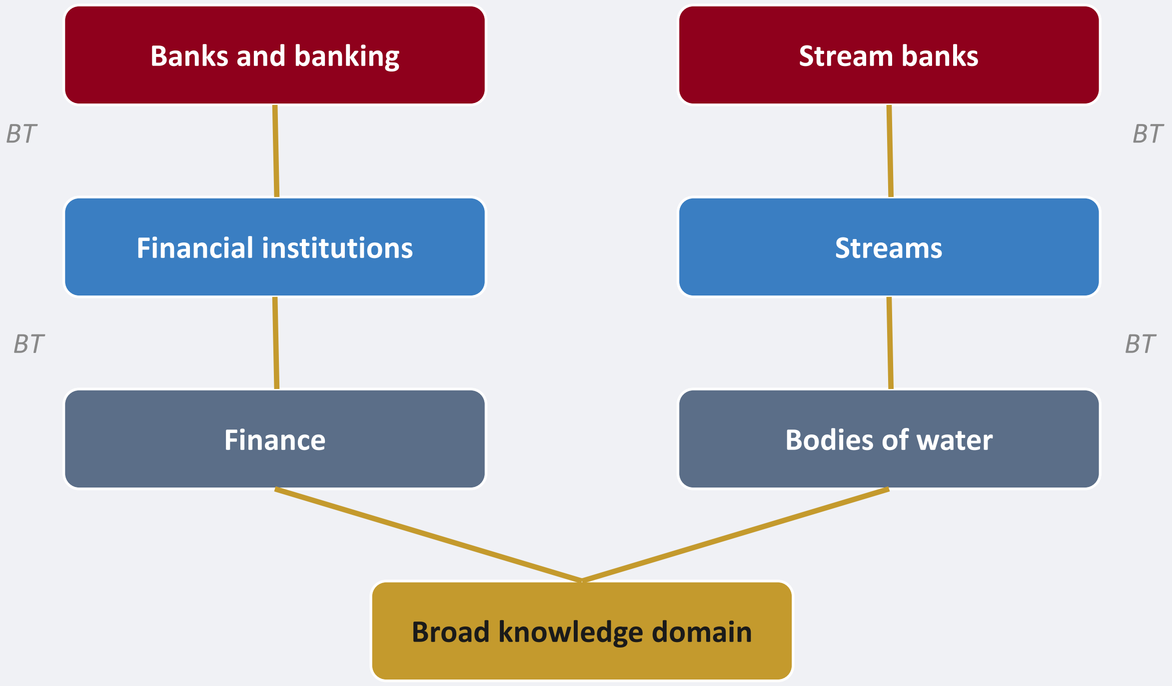
Source: id.loc.gov — illustrative adaptation

LCSH as the Test Case

Formalizing LCSH's hierarchical structure as a directed acyclic graph

$$\texttt{hier\_distance}(h_1, h_2) = d(h_1, c^*) + d(h_2, c^*)$$

Refer to the LCSH Graph Poster →



No common ancestor within D ≤ 3 steps → same_nbhd = FALSE
No common ancestor at all → same_nbhd = FALSE, hier_distance = ∞

Independent Variables

`distance_bert`

Continuous

BERT cosine distance between contextual embeddings of the two word senses.

`same_heading`

Binary (0/1)

Do both senses map to the same authorized LCSH heading?
The primary categorical predictor. Core test of H2 — cognitive warrant.

`hier_distance`

Continuous integer

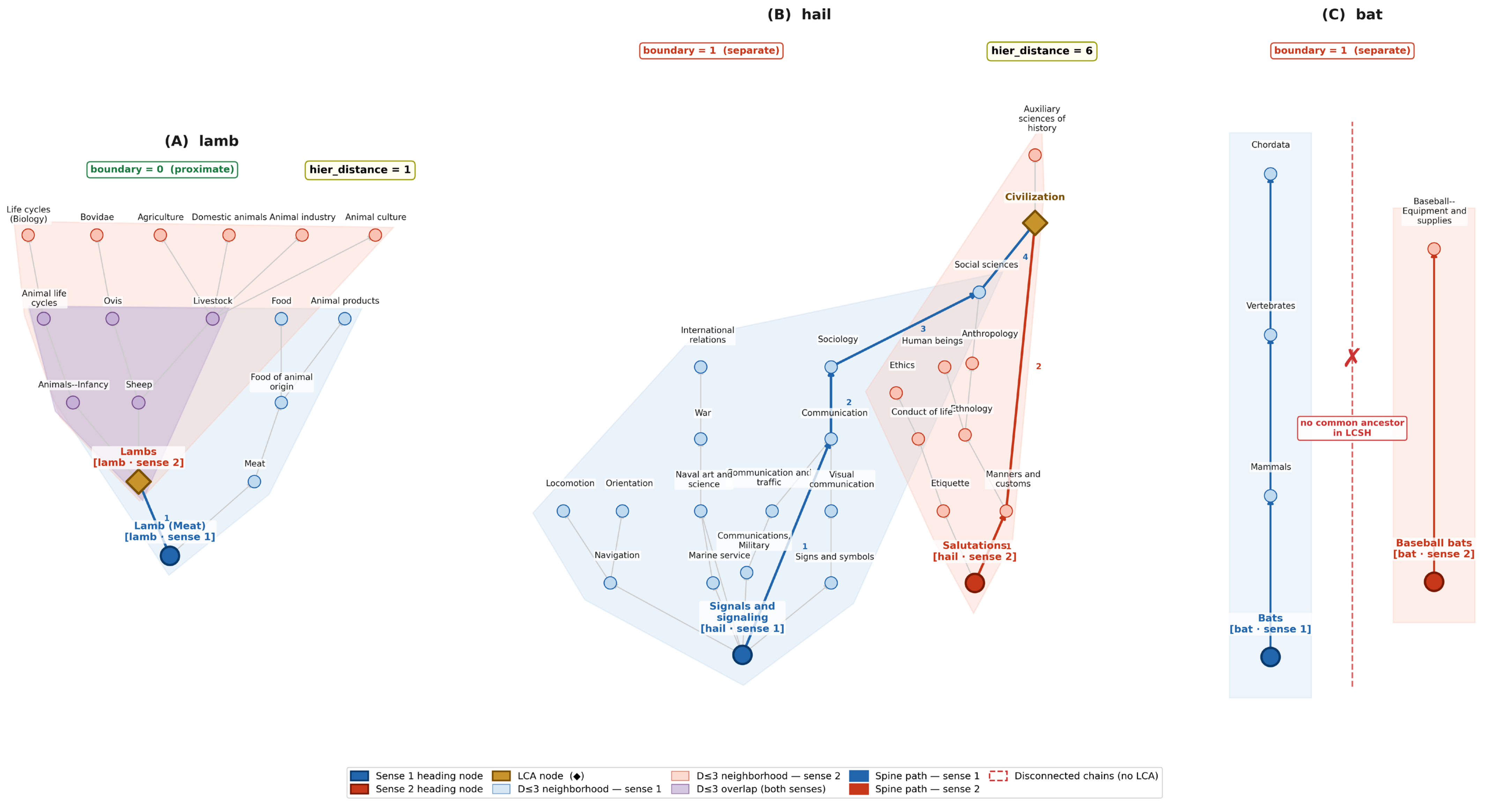
BT-hop path length between the two headings via LCA.
Extracted from MADS/RDF bulk download using NetworkX.

`same_nbhd`

Binary (0/1)

Do both headings share a BT ancestor within D ≤ 3 steps?
Preregistered threshold. Enables continuous hier_distance within neighborhoods.

LCSH Hierarchical Subgraph Structure



LCSH hierarchical subgraph structure for three polysemous words. Each panel shows a directed acyclic graph in which edges point from narrower (child) to broader (parent) term. Colored spines trace the route from each sense heading to their Lowest Common Ancestor (LCA, ◆ gold diamond); edge labels show cumulative hop distance from the sense heading. Shaded regions show the D≤3 neighborhood of each sense. hier_distance = total edges separating the two senses via their LCA. lcsh_boundary_binary: 0 = same hierarchical branch (proximate); 1 = separate branches or no common ancestor.

Processing of MADS/RDF XML into a directed graph

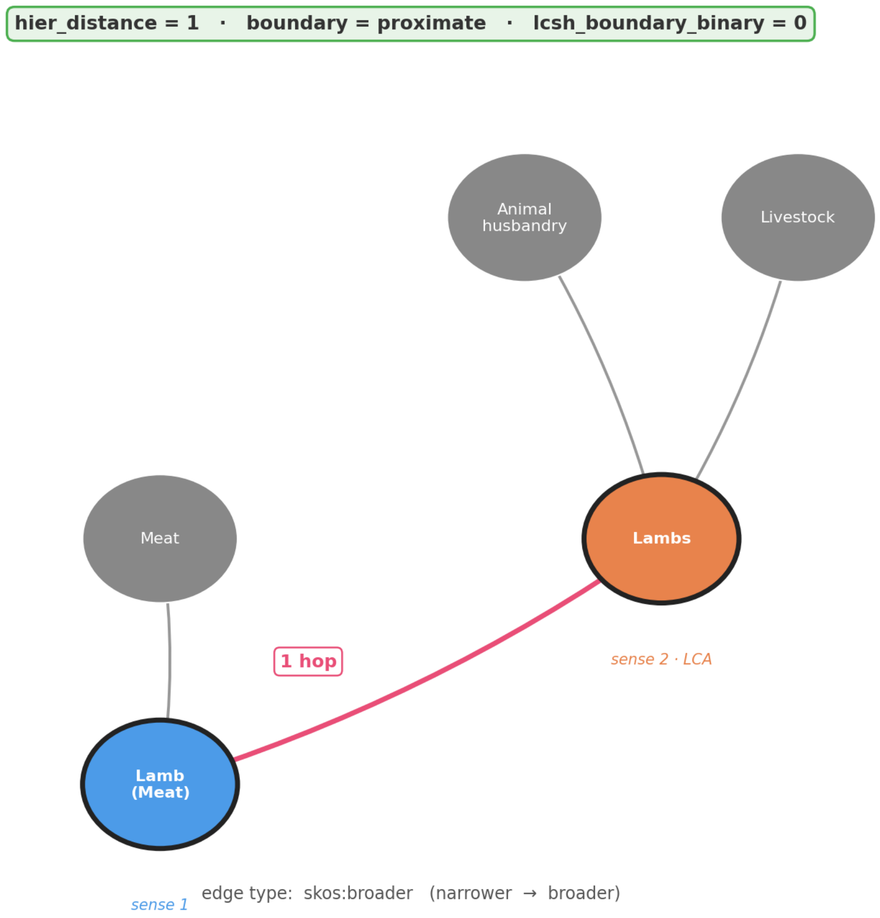
A Raw MADS/RDF XML (one of ~330,000 records)

```xml
<rdf:RDF xmlns:rdf ="...rdf-syntax-ns#" ...
  xmlns:madsrdf ="http://www.loc.gov/mads/rdf/v1#" >

  <madsrdf:Topic rdf:about ="…/sh85074167" >
    <madsrdf:authoritativeLabel xml:lang ="en" >
      Lamb (Meat)
    </madsrdf:authoritativeLabel>
    <bflc:marcKey> 150 $aLamb (Meat) </bflc:marcKey>
    <madsrdf:isMemberOfMADSScheme rdf:resource ="…/subjects" />

    <madsrdf:hasBroaderAuthority>
      <madsrdf:Topic rdf:about ="…/sh85074185" >
        <madsrdf:authoritativeLabel xml:lang ="en" >
          Lambs
        </madsrdf:authoritativeLabel>
      </madsrdf:Topic>
    </madsrdf:hasBroaderAuthority>

    <skos:broader rdf:resource ="…/sh85074185" />

  </madsrdf:Topic>
</rdf:RDF>
```

2,400+ namespace prefixes · nested blank nodes · ~330 K records · 2.6 GB

parse + extract

B Parsed graph input (NetworkX DiGraph)



Research Design as Demonstration — and Structural Validation

Cognitive warrant is empirically testable — and preliminary data confirm the framework

Experimental Design

1 Primed Sensibility Judgment

DV: log(RT) — implicit processing difficulty

Prime sentence establishes sense context → target sentence → keyboard response (sensible?)
Key: does crossing an LCSH heading boundary increase RT, controlling for BERT distance?

2 Heading Appropriateness Rating

DV: Δr = rating of target − rating of within-category foil

Two sentences (sense context) → rate 4 LCSH headings (1–7).
BERT-matched foils isolate categorical boundary effect from semantic distance.

Likelihood Ratio Test Model Sequence (Experiment 1)

M_0 Unconditional model → + distance_bert [H1] → + same_heading [H2] → quadratic interaction [H3]

Preliminary Structural Validation

From 150 LCSH-mapped stimulus pairings (25 words × 6 pairs):

H2 confirmed $p < 0.0001$
LRT χ^2= 27.02, df=1

Odds ratio (B) 19.6×
Same heading makes same-sense. 20× more likely controlling for distance

CI-bar 0.13 bits
Positive expected cognitive information

Warrant efficiency 18.0%
Residual ambiguity resolved by LCSH categorical structure

CI(d) peak Medium bin
Consistent with H3 as mathematical consequence

Revision priority dinner
pCI = −2.45; same heading assigned to distinct senses

Example stimuli from Trott and Bergen (2023)

Each context sentence establishes a specific word sense through natural usage.

Target Word Stimuli (Librarian validation, Experiment 1, and Experiment 2)
Sense 1 sentences (S1a, S1b) and Sense 2 sentences (S2a, S2b) are the prime–target pairs used to instantiate each word sense.

book (Noun ▪ Polysemy)
Sense 1:
S1a. He had a best-selling **book**.
S1b. He had a popular **book**.
Sense 2:
S2a. He had a heavy **book**.
S2b. He had a leather-bound **book**.

lap (Verb ▪ Homonymy)
Sense 1:
S1a. She **lapped** the milk.
S1b. She **lapped** the water.
Sense 2:
S2a. She **lapped** the runners.
S2b. She **lapped** the competitors.

Instance [BIBFRAME] or Manifestation [FRBR] sense: “He had a best-selling book.”
Item sense: “He had a leather-bound book.”

Filler Sentences (Experiment 1 Only)

One sentence makes grammatical sense:

He liked to play **golf**.
He liked to swim **golf**.

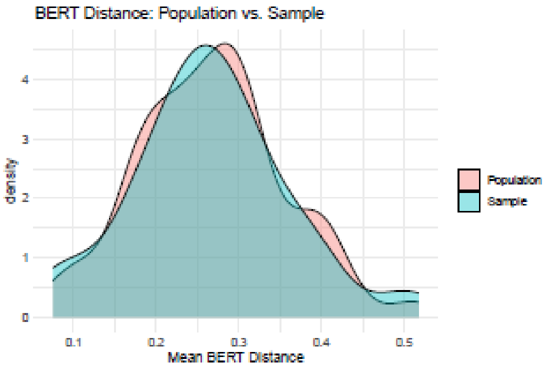
Neither sentence makes grammatical sense:

He **packed** the freedom.
He **packed** the ocean.

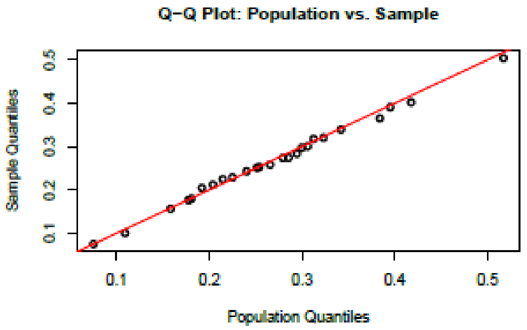
Distributional comparison of stratified random sample with full stimulus set

Table 4.1: Distributional Comparison of BERT Semantic Distance: Population vs. Sample

Metric	**Population** ($N = 104$)	**Sample** ($n = 25$)
Mean	0.267	0.265
Median	0.266	0.258
SD	0.088	0.095
Min	0.075	0.075
Max	0.517	0.504
Skewness	0.248	0.271



(a) Kernel density estimates of mean BERT distance for population and sample.



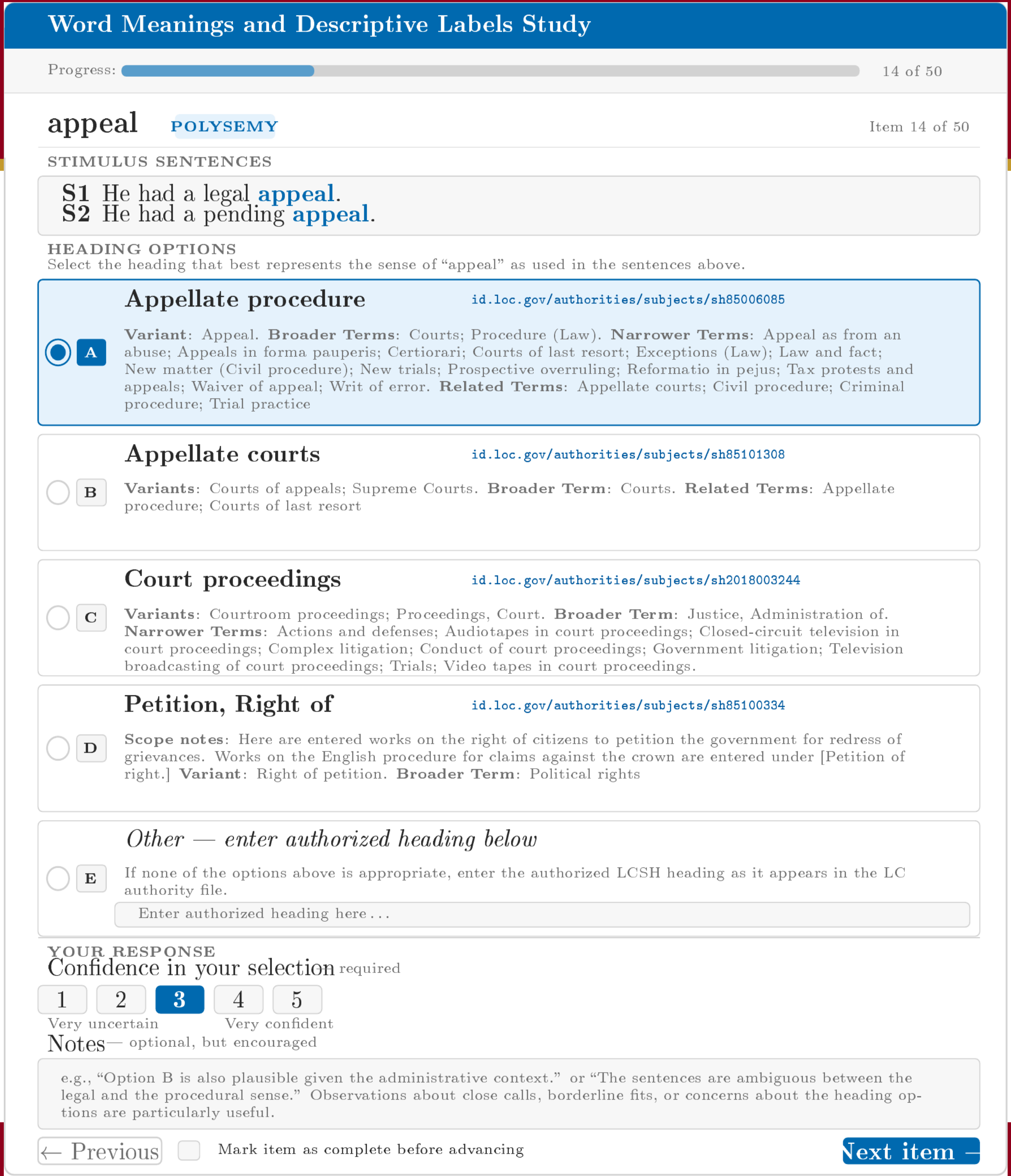
(b) Quantile-quantile plot: population vs. sample.

Figure 4.1: Distributional equivalence of stratified random sample ($n = 25$) to source population ($N = 104$) on mean BERT cosine distance.

LCSH candidate evaluation rubric

1. **Semantic Fit**: the degree to which the LCSH preferred label and scope note correspond to the intended meaning of the stimulus word sense as defined by contextualized sentence usage.
2. **UF Variant Coverage**: the extent to which the authority record contains lexical variants (*Use For* fields) corresponding to the stimulus word or close synonyms.
3. **Hierarchical Placement Consistency**: the degree to which the broader and narrower term relationships of the candidate heading align with the conceptual domain implied by the stimulus sense.
4. **Specificity (Conceptual Granularity)**: the degree to which the heading captures the intended level of conceptual specificity without being overly broad or overly narrow.
5. **Cataloging Applicability**: the extent to which the heading functions as a valid topical subject heading for bibliographic indexing rather than a structural or subdivision element.
6. **Boundary Correspondence**: the extent to which the candidate heading pre-serves the intended categorical sense distinction relative to other senses of the same word.

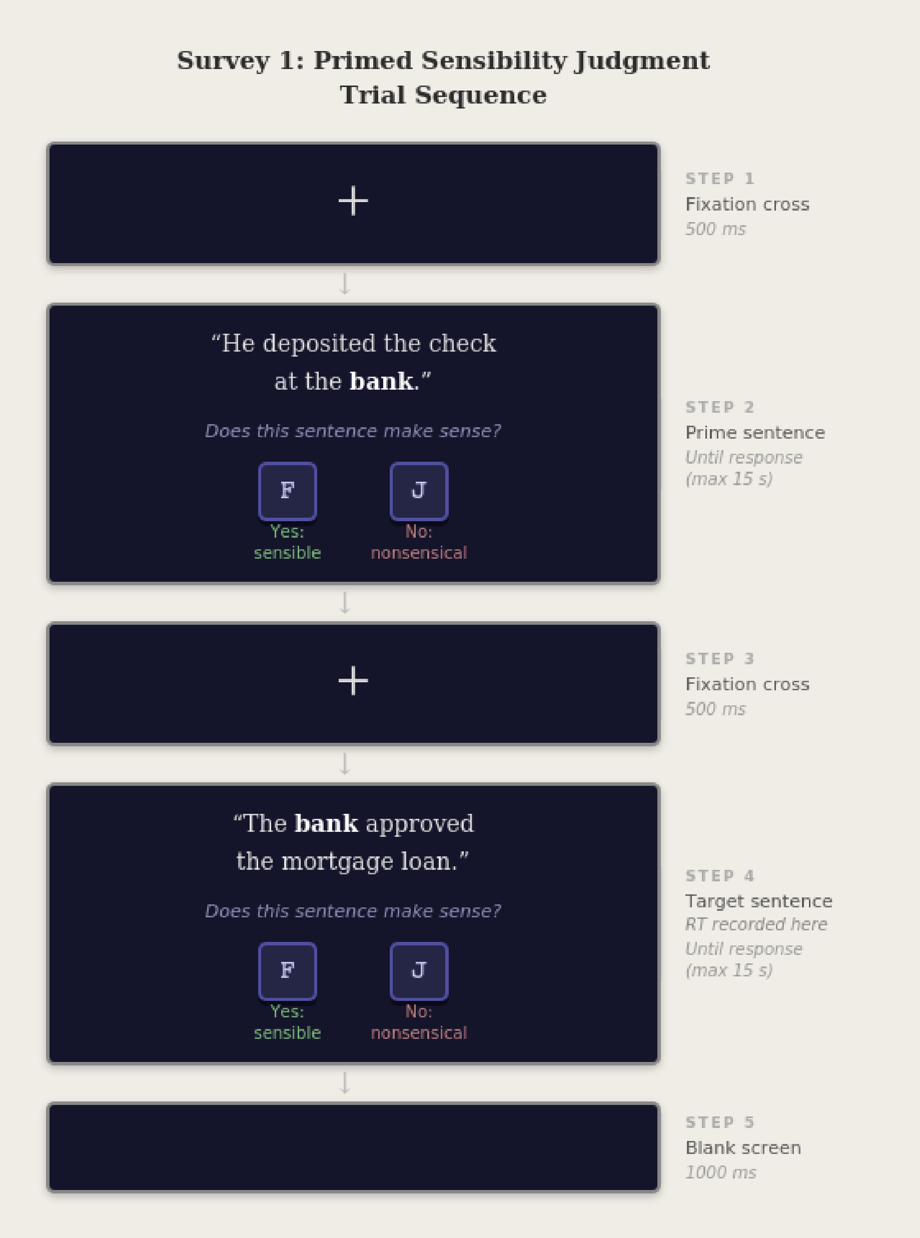
Mock-up of the Qualtrics-based librarian evaluation interface

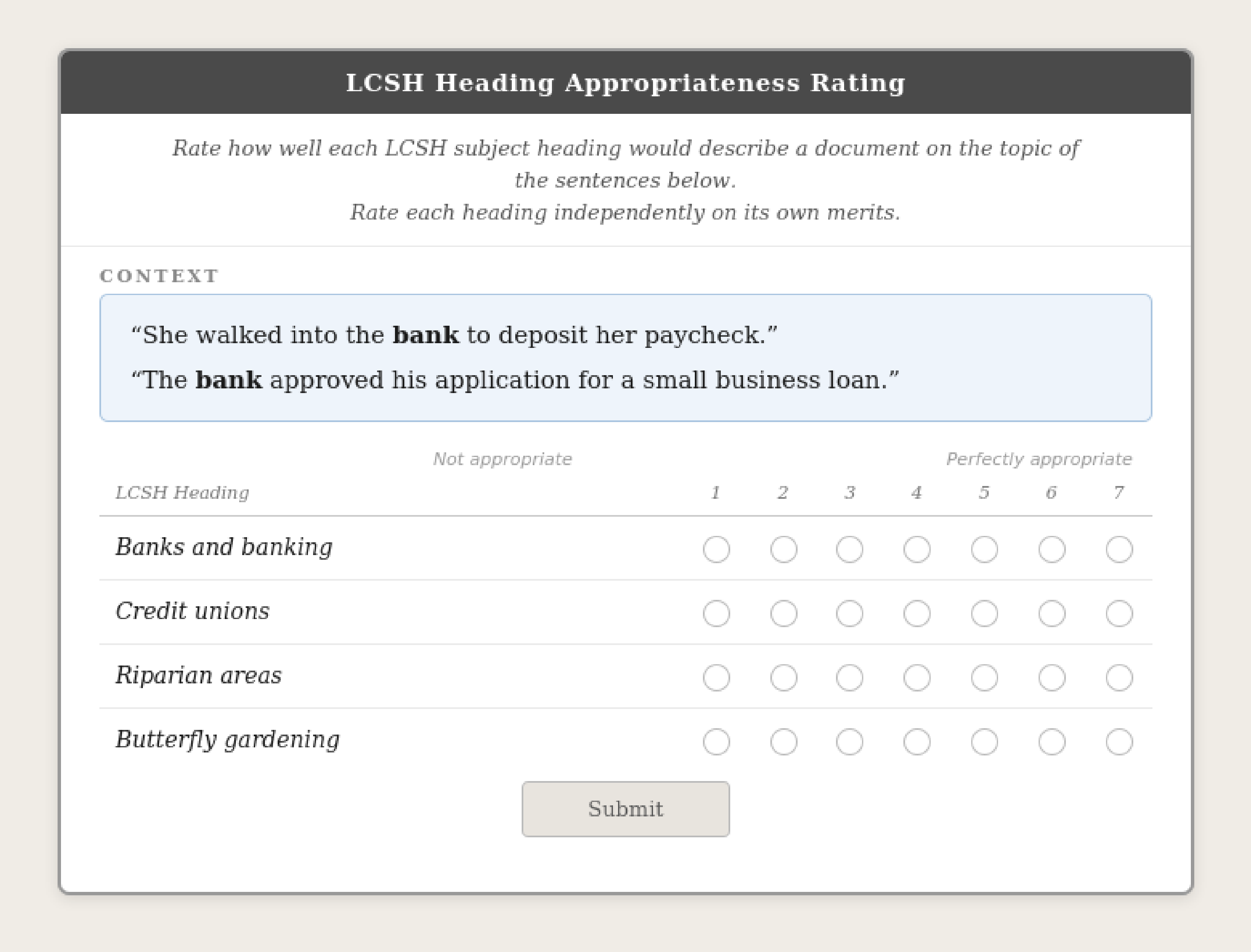


Documentation of LCSH mapping decisions

Sense #	Word	Word Sense ID	Selected LCSH Heading	LCSH URI	# UF Variants	UF Variants	Topical Subdivision?	Selection Justification
31.	lap	lap_1	Drinking behavior	http://id.loc.gov/authorities/subjects/sh85039580	**0**	(none)	**No**	The lapping/drinking-with-tongue sense is most precisely captured as a behavior related to liquid consumption. Drinking behavior is the closest available heading for the act of consuming liquid (though in LCSH it primarily covers human alcohol-related behavior). Tongue (Candidate 2) is the anatomical organ rather than the action; zero UF entries for Drinking behavior is a noted limitation, and this represents a genuine vocabulary gap in LCSH for animal drinking behavior.
32.	lap	lap_2	Racing	http://id.loc.gov/authorities/subjects/sh85110265	1	Races (Sports)	**No**	The sense of lapping a competitor in a race is subsumed under Racing broadly. UF entry "Races (Sports)" confirms. Automobile racing (Candidate 2) is too domain-specific; Track and field (Candidate 3) is limited to athletics; Sports—Rules (Candidate 4) is about rule systems, not the competitive act. Racing is the broadest applicable heading.

Mock-up of the interface participants would encounter





Generation of Experiment 2 stimuli

Experiment 2 makes the categorical judgment explicit. Each trial presents two stimulus sentences simultaneously. Participants rate four LCSH headings on a 1–7 appropriateness scale:

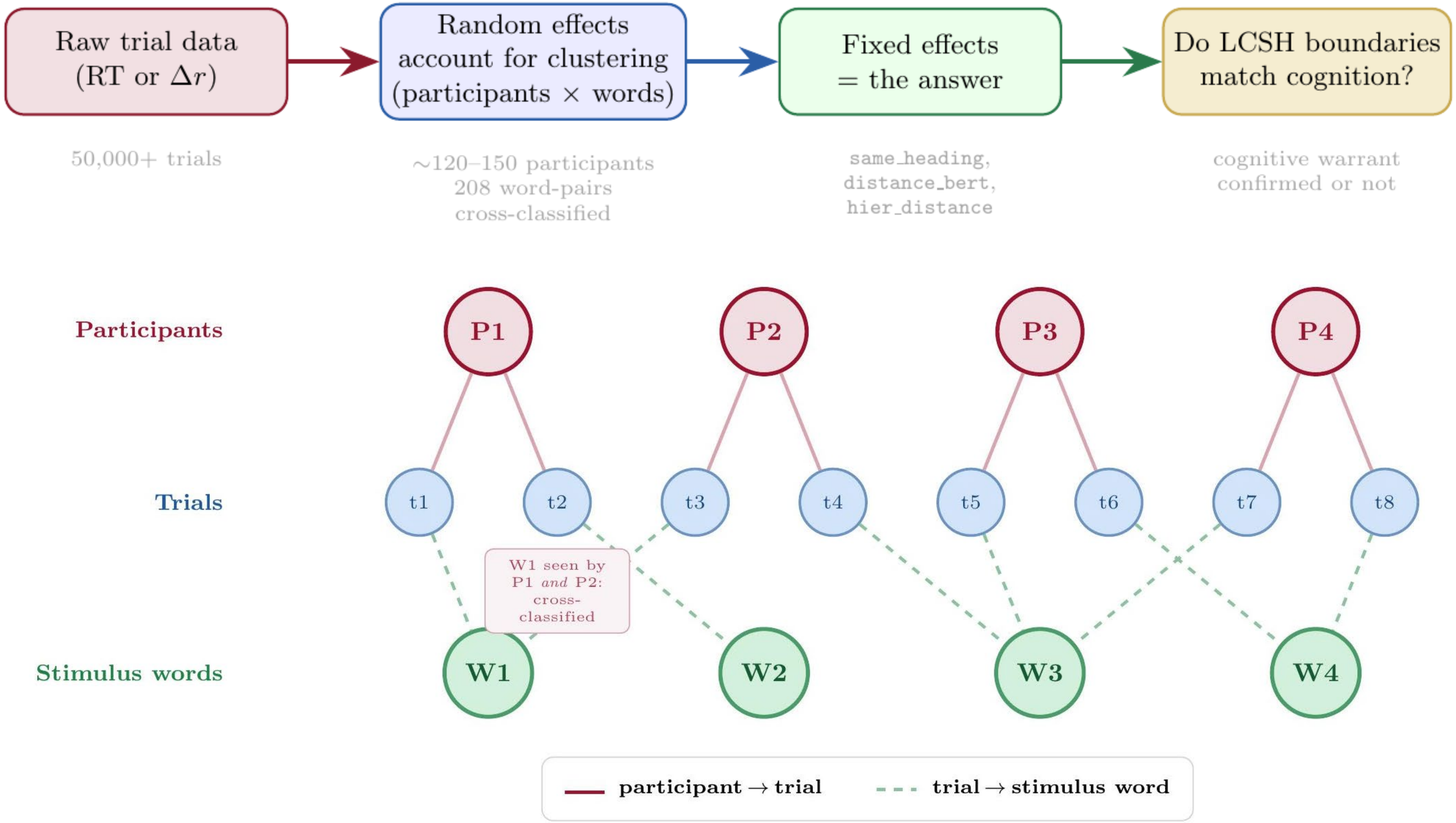
1. **Target heading**: The authorized LCSH heading mapped to the target sense.
2. **Within-category foil**: A different LCSH heading in the same hierarchical neighborhood (`same_nbhd` = 1) with `distance_bert` matched to the cross-category foil (± 0.05).
3. **Cross-category foil**: An LCSH heading from a different neighborhood (`same_nbhd` = 0) with `distance_bert` matched to the within-category foil.
4. **Attention check**: A clearly irrelevant heading.

BERT-distance matching between foil types is the critical design feature: it isolates the effect of categorical boundary position from semantic distance, attributing any rating difference specifically to categorical structure.

The primary dependent variable is the discrimination score: $\Delta r = r_{target} - r_{within-category\ foil}$

The statistical approach

Cross-classified hierarchical linear model



LCSH Is the Test Case — The Framework Is the Contribution

Cognitive warrant applies to any controlled vocabulary with categorical boundary structure

Applicable Knowledge Organization Systems

Library & Bibliographic

- LCSH (primary test case)
- Medical Subject Headings (MeSH)
- Art & Architecture Thesaurus (AAT)
- FAST (Faceted Application of Subject Terminology)

Biomedical & Clinical

- Unified Medical Language System (UMLS)
- National Cancer Institute Thesaurus
- SNOMED CT

Multilingual Systems

- Rameau (Bibliothèque nationale de France)
- NDLSH (Japan National Diet Library)
- UNESCO Multilingual Thesaurus

Computational & Linked Data

- DBpedia ontology
- Schema.org
- Wikidata

The Cross-Vocabulary Question

Warrant efficiency profiles estimated from the same stimuli constitute a principled, distance-stratified comparison of the cognitive value of each system's categorical structure:

$$\eta\_LCSH(d) \quad vs. \quad \eta\_MeSH(d) \quad vs. \quad \eta\_BERT\text{-}cluster(d)$$

If BERT-cluster η > LCSH η

Evidence for reconsidering how AI-assisted cataloging tools are evaluated and deployed.

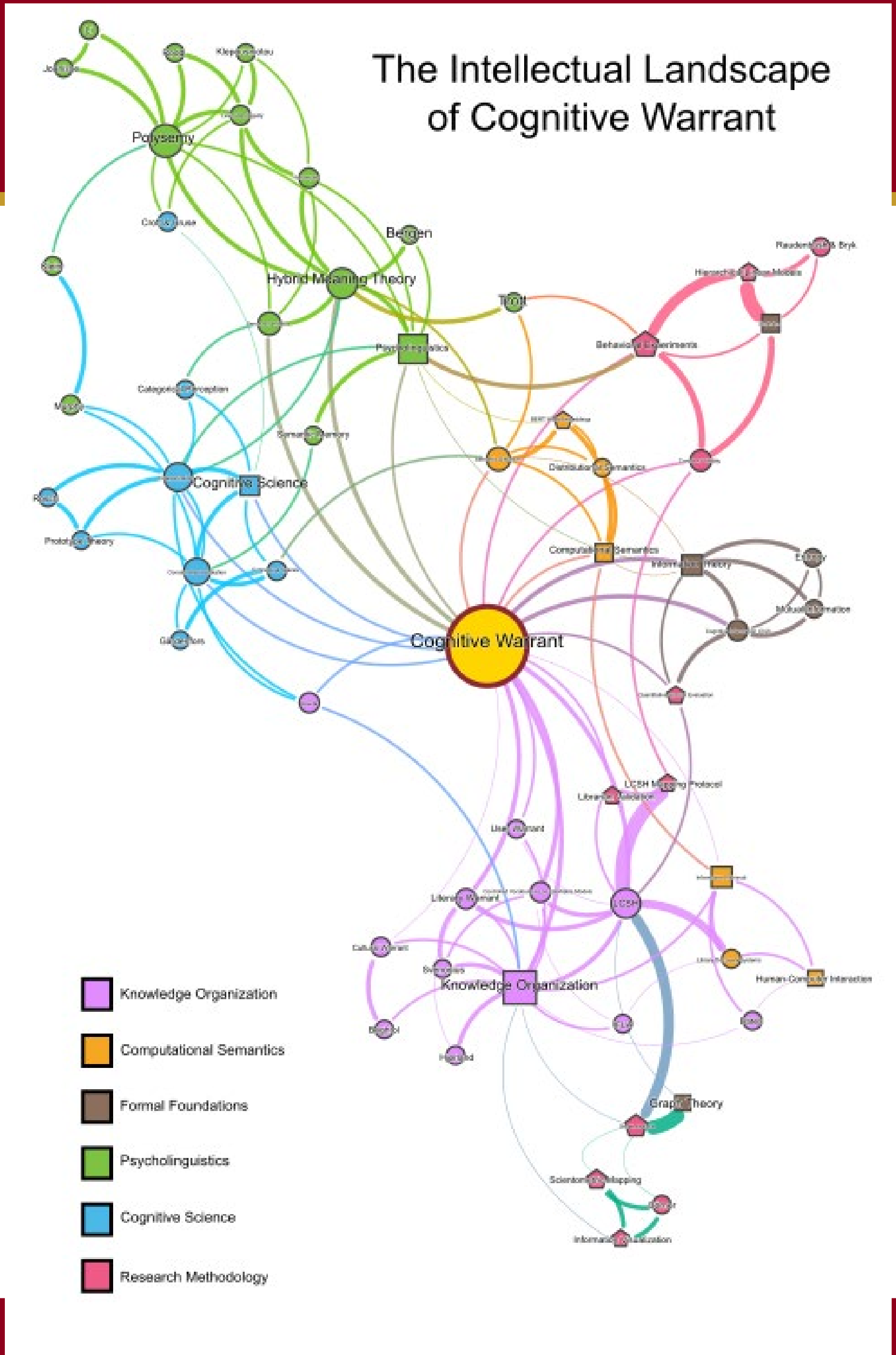
If LCSH η ≥ BERT-cluster η

A century of accumulated editorial judgment tracks cognitive structure more reliably than distributional optimization alone.

Regardless of outcome

The framework generates actionable evidence and positions KOS research to engage empirically with cognitive systems.

The intellectual landscape of cognitive warrant



Where This Goes

1 Standalone Theory Paper

Cognitive warrant as a general KOS evaluation criterion, independent of LCSH. Target: Journal of the American Society for Information Science and Technology or Knowledge Organization.

2 Full Behavioral Study

N ≈ 200 participants. IRB approval secured. Full LRT sequence testing H1–H3. CI-bar and pCI derived from actual RT and discrimination scores.

3 Cross-Vocabulary Comparison

MeSH as the priority. Same stimulus materials, same pipeline. $\hat{\eta}$_LCSH vs $\hat{\eta}$_MeSH as the first direct empirical comparison of cognitive value across major controlled vocabularies.

4 Doctoral Research Program

The framework, the methodology, and the most consequential questions are now precisely statable. Cognitive warrant as a research program, not just a capstone.

"

Knowledge organization systems do not merely describe the world — they impose structure on how users encounter it.

Cognitive warrant is the principle that makes those claims testable — and positions knowledge organization research to engage empirically with the semantic processing and cognitive structure of the users library discovery is designed to serve.

Nicholas W. Meister
MLIS · University of Denver · June 2026

Acknowledgement of my “study buddies”

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Thank you!